

Ink Harmony: An AI- and VR-Enhanced System for Calligraphy Education

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ABSTRACT

This work introduces an assistive system designed to deepen the engagement with the art of calligraphy and Chinese culture. Users input the text they wish to learn, which is then processed by an AI framework to create a thematic photo. This photo forms part of a virtual environment, establishing an environment that contextualizes the calligraphy experience. Simultaneously, the system communicates with a robotic arm, guiding the user's hand to improve writing technique and engagement with the art form. This combination of AI, virtual reality, and robotic assistance paves the way for interactive learning and appreciation of calligraphy.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**; • **Applied computing** → *Media arts*; • **Computing methodologies** → Virtual reality; Artificial intelligence.

KEYWORDS

calligraphy, education technology, virtual reality, artificial intelligence, interaction design, media arts

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1 INTRODUCTION

As society undergoes modernization and globalization, the pace of life accelerates, and the influence of social media fragments information, fewer individuals dedicate themselves, as ancient literati or critics did, to the profound study of such rich poetic traditions and literary theories. The humanities, with a focus on the spiritual aspect of humanity, face the significant question of how to effectively convey the allure of traditional culture to the public in a manner that evokes mental and psychic resonance and creates an enchanting refuge amidst urban landscapes dominated by concrete

and steel, especially the most traditional poetic sentiments and the Rhyme of Calligraphy. Fortunately, we live in an era of advanced technology and rapid development, where artificial intelligence continuously challenges the boundaries of human cognition, and VR technology allows us to transcend ourselves and see a different world [Disguise 2023; Huang et al. 2023; Nametag Films 2020].

Relevant work has incorporated emerging generative and immersive techniques into content creation and understanding [Lin et al. 2023; Waite 2023; Yu et al. 2021], and some work has discussed the AI-induced paradigm shift in the production industry [Song et al. 2023; Waite 2020; Yip 2022]. However, few combine AI-generated tools and immersive VR environments to facilitate calligraphy education. Hence, we present Ink Harmony, an innovative and integrated assistant system that aims to revolutionize the traditional method of calligraphy practice by seamlessly combining AI, virtual reality, and robotic assistance. This system is designed to deepen the engagement with the art of calligraphy and the cultural identity of China, offering users a comprehensive and interactive learning experience.



Figure 1: A user experiencing Ink Harmony in a VR headset.

2 DESIGN AND IMPLEMENTATION

Ink Harmony begins with users inputting the text they wish to learn, typically an ancient poem or literary excerpt. This text serves as the cornerstone of their calligraphy journey. The system then utilizes an AI framework [Wang et al. 2023] to analyze the textual input, extract its thematic and emotional essence, and generate a coherent and engaging immersive environment to support the understanding and creation of participants. This framework is crucial

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because it ensures that the environment is tailored to the specific characteristics of the text, making the learning experience more personalized and meaningful.

The AI framework first performs a sophisticated linguistic analysis, identifying key emotional and thematic elements within the text. This data is then transformed into a panoramic image, a visual representation that encapsulates the spirit and context of the text. The thematic visual is not merely an illustration, but a profound interpretation of the text, reflecting its deeper meanings and nuances. This image serves as the visual foundation for the VR environment, bridging the gap between textual analysis and visual representation. We also added some 3D models, including a set of Chinese tables and chairs and writing brushes from the Unity Asset Store to the VR environment.

The participants could engage in calligraphy practice within the VR environment, holding the physical brush sticks and following the guidance of the robotic arm. This combination creates a rich multisensory learning experience. The immersive VR setting enhances the sense of presence and interaction, while the precise movements of the robotic arm ensure accurate stroke replication. This method not only makes the learning process more enjoyable but also more effective, as users can practice and refine their skills in a supportive and engaging environment.

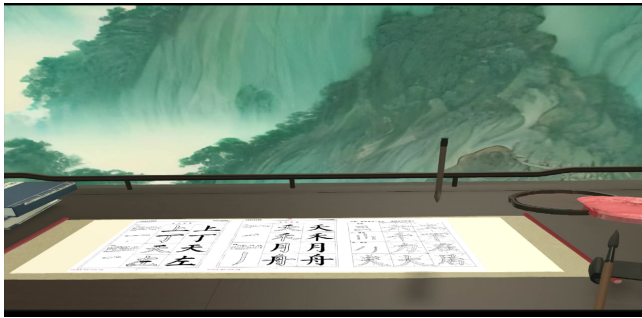


Figure 2: Viewing the virtual environment in Unity.

This seamless blend of AI, VR, and robotic assistance paves the way for a new era of interactive learning and appreciation of calligraphy. It not only transforms the act of learning into a rich, multisensory experience but also opens the door for individuals of all culture backgrounds to access and engage with the art of calligraphy and Chinese culture. Through this integrated system, we explore the boundless possibilities at the intersection of technology, culture, and education, ushering in a transformative era for traditional art forms.

3 DISCUSSION AND FUTURE WORK

Ink Harmony represents a novel assistive system for calligraphy education, leveraging an AI model to visualize the deep meanings embedded in ancient Chinese poems by creating an immersive skybox environment. This approach significantly improves the engagement and immersion of the calligraphy practice experience, greatly contributing to the dissemination and preservation of the Chinese traditional culture.

However, the performance of calligraphy writing is influenced by several factors that require further attention and improvement. The first factor is the robot arm, which inevitably experiences vibrations that can cause strokes to be less smooth. Additionally, we cannot incorporate more complex stroke transformations such as slants and apostrophes due to the limitation of the degree of freedom (DoF) of the robot arm. The second factor is the VIVE Tracker. For accurate positioning, the tracker must remain unobstructed at all times. Any obstruction, even momentary, can affect the real-time precision and overall performance of the system. Thirdly, the current robotic arm writing lacks the nuanced variability inherent in human writing behavior. Emulating human writing dynamics, including variable speed and subtle movements, is vital to achieving a more realistic and effective calligraphy practice. Furthermore, font stylization is important to accommodate different user preferences and enhance the personalization of calligraphy practice.

Addressing these challenges will be crucial for the future development of Ink Harmony. By tackling these issues, we continuously refine the system to enhance its capability to provide an engaging and effective calligraphy learning experience.

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